

Self-Improving Customer Intelligence Engine

Turn every customer question into product & marketing intelligence.

For: Boldr Supply Co. · Premium titanium watch brand, Singapore

Team AlphaBeta · Tejas Chavan + Ayush K Pacheriwala

Who is Boldr — and why this matters

BOLDR SUPPLY CO.

Premium titanium watch brand

- **Singapore HQ.** Two flagship models (Expedition, Journey) + a sold-out limited edition.
- **Channels:** email, chat, WhatsApp, Instagram DM.
- **CS team:** 3 people sharing one Gmail inbox. Fully manual.
- **Stack:** Shopify, Google Workspace, Slack — no automation today.
- **Customer base:** global, with gifters, enthusiasts, and health-conscious buyers.

THE PRESSURE TODAY

70

tickets analysed

Nov 2025 — May 2026

33%

are knowledge gaps

no system to capture or learn

8 min

to draft each reply

across 4 channels, manually

Boldr's most valuable underused dataset

CATEGORY

Customer support automation + Market intelligence for SMEs

PROBLEM STATEMENT

Boldr's 3-person CS team is drowning in repetitive enquiries, while every customer question — a real buyer signal — is discarded the moment the reply is sent.

Repetition tax

BPA-free, engraving pricing, strap compatibility — the same answers re-typed daily.

67% of tickets are KB-answerable but drafted from scratch every time.

Knowledge that walks out the door

Resolved tickets stay in Gmail. Nothing flows back into the FAQ, SOP, or product pages.

Arabic engraving was flagged as a gap — but the rate card already supports it.

Marketing is blind to real demand

Sustainability, child safety, multi-script engraving — asked weekly, surfaced never.

5 product-page content gaps surfaced from a single month of tickets.

Seven steps. Every ticket teaches the system.



↪ Steps 5 → 2 form the self-improving loop: resolved gaps become new KB entries the engine can answer next time.

FOR CUSTOMERS

Faster, on-brand replies — no hallucinated answers, ever.

FOR CS TEAM

Review-and-approve instead of draft-from-scratch. ~80% time saved on KB tickets.

FOR MARKETING

Monthly brief surfaces unmet demand + product-page gaps with persona tags.

Built once, ownable by a 3-person team

THE STACK

Orchestration	n8n (visual canvas, self-host or cloud)
AI model	Qwen3-32B via FPT AI Factory
Knowledge Base	Google Sheets + vector embeddings
Approval gate	Google Sheets + Slack pings
Outbound	Gmail API (post-approval only)
Ticket source	Webhook from email / chat / DM

7 BUYER PERSONAS · TAGGED IN PARALLEL AT STEP 1b

Derived from real ticket data — validated live in n8n.

● health_conscious	BPA, nickel, hypoallergenic, dye safety
● gifter	engraving, gift wrap, occasions
● enthusiast	specs, movement, compatibility, LEs
● niche_buyer	magnetic, lume, depth, niche certs
● owner_aftercare	servicing, repairs, warranty, polish
● prospect	price match, stock, comparisons
● transactional	shipping, customs, tracking, refund

Two real tickets, end-to-end

DEMO A • ANSWERABLE FROM KB • TKT-1048

Vikram Allen • Chat

“I’m buying this for my young daughter and want to make sure the strap is BPA-free. Can you confirm?”

Step 1 — Classify	materials_safety • health_conscious (high)
Step 2 — KB search	FAQ p.1 + Product Reference • conf. 0.97
Step 3 — Draft reply	Brand-voice reply • queued for human approval
Outcome	Sent in ~90s instead of ~8 min

✓Validated in live n8n execution

DEMO B • KNOWLEDGE GAP • TKT-1013

Victoria Singh • Chat

“Do you offer carbon-neutral shipping? I try to offset my purchases where possible.”

Step 1 — Classify	knowledge_gap • niche_buyer
Step 2 — KB search	No match • conf. < 0.75
Step 4 — Flag gap	Logged, routed to CS • no hallucination
Step 5 — KB draft	After CS reply, drafts new entry for 1-click approval

↪ Gap becomes a KB entry — system never asks this twice

Four deliverables — already generated for May

1

Drafted replies

Brand-voice, KB-cited, human-gated.

5 samples • human approval
queue

*phase5_outputs/
01_sample_drafted_replies.md*

2

Knowledge gap log

Every unanswered question, with
persona + channel context for CS.

14 entries • 79% resolved

*phase5_outputs/
02_knowledge_gap_log.md*

3

Weekly theme clusters

Group novel questions by theme.
Trend flags: new, recurring,
escalating.

4 clusters • May 5-11

*phase5_outputs/
03_weekly_theme_clustering_repor
t.md*

4

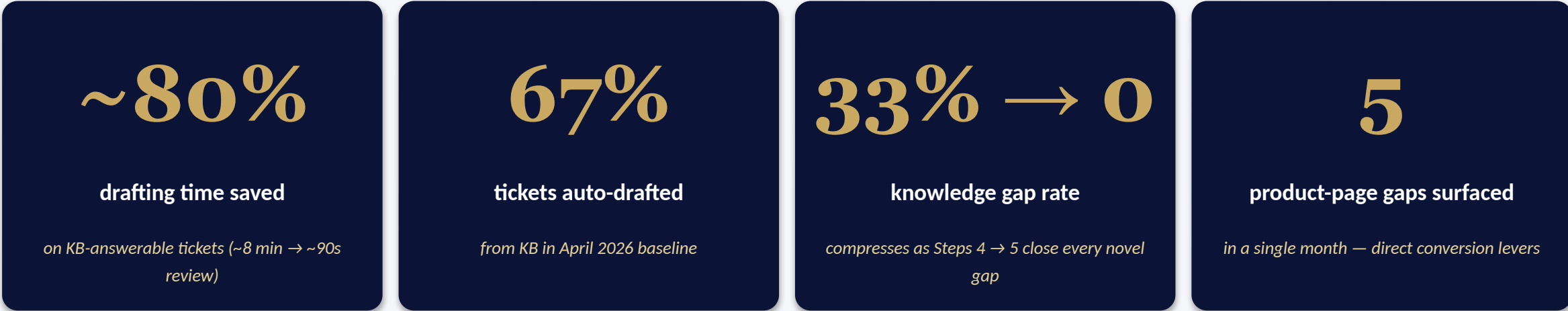
Monthly marketing brief

Top themes, persona mix, product-
page gaps, content
recommendations.

5 content gaps • April 2026

*phase5_outputs/
04_monthly_marketing_brief.md*

Time saved, revenue surfaced, knowledge compounded



OPERATIONS

Recover ~30 CS hours per month

Review-and-approve workflow replaces draft-from-scratch on routine queries. Three-person team reclaims a full working day every week — without hiring.

REVENUE

New marketing intelligence surface

Monthly brief surfaces unmet demand (multi-script engraving, child safety, sustainability) — each is a copy / FAQ change that lifts conversion on pages customers already visit.

RETENTION

Loyalty signal from owner_aftercare cluster

17% of April tickets are existing owners asking about servicing. Faster, consistent answers protect lifetime value and feed an upsell path to Premium Service (SGD 220).

Under SGD 50/month — recovers ~30 CS hours

MONTHLY COST BREAKDOWN • BOLDR-SCALE OPERATION

Component	Setup	Monthly	Notes
n8n (workflow host)	—	\$0	Free tier covers Boldr volume
Qwen3-32B (FPT AI Factory)	—	~\$15-25	OpenAI-compatible API, sponsor credits during competition
Embeddings (KB vectors)	—	~\$2	One-time index + drift refresh
Google Sheets / Slack / Gmail	—	\$0	Existing Boldr stack
Vector store (Supabase pgvector)	—	\$0	Free tier sufficient at this scale
Prompt engineering (one-time)	Done	—	Shipped — see repo for prompts

TOTAL **≈ SGD 35-50** *all-in, monthly*

RETURN ON INVESTMENT

~30 hrs

of CS time recovered per month

- **Scales free**
Sub-linear cost — most growth absorbed by free-tier headroom.
- **Sponsor credits**
FPT AI Factory covers inference cost during competition + early pilot.
- **No headcount needed**
Replaces a hire, not a person — 3-person CS team keeps full ownership.
- **Commercial fallback**
If FPT credits end, Qwen3-32B at OpenAI-compat rates stays cheaper than GPT-4o-mini for this volume.

Responsible AI is the architecture, not a feature



No auto-send

Every AI-drafted reply enters a Google Sheets 'Pending Approval' queue. CS clicks Approve / Edit / Reject before Gmail sends anything.



No hallucination

If KB confidence < 0.75, the engine never drafts an answer. The ticket routes to a human queue with full context — silence is preferred to invention.



No auto-publish to KB

Step 5 drafts a proposed KB entry from a resolved gap. Team lead must 1-click approve before the entry is indexed and used for future replies.



PII boundary

Tickets stay in Boldr's Google Workspace. Only message body is sent to the LLM; order IDs are extracted locally and not forwarded where avoidable.



Pricing safety

Where the SOP conflicts with the rate card (e.g. SGD 35 vs SGD 40 for engraving), the rate card is authoritative — caught and codified in the KB.



Audit trail

Every classification, KB match, draft, and approval is logged to Google Sheets with timestamp. CS can trace any decision the engine made.

Live in n8n, public repo, validated on real tickets

BUILD STATUS • n8n @ tejasdotchavan.app.n8n.cloud

Webhook trigger	LIVE
Step 1 — Ingest & Classify	LIVE
Step 1b — Persona tagger	LIVE
Step 2 — KB Search routing	LIVE
Step 3 — Draft reply (brand voice)	LIVE
Step 4 — Flag knowledge gap	LIVE
Step 5 — Auto-draft KB entry	WIRING
Step 6 — Weekly clustering	WIRING
Step 7 — Monthly brief	DRAFTED

Validated live: Vikram Allen BPA ticket → health_conscious ✓

ARTEFACTS FOR JUDGES



- Public repo `tejasdotchavan/e27-boldr-challenge-AlphaBeta`
- Architecture doc `workflow_architecture.md` (24KB)
- Data analysis `data_observations.md` (14KB)
- Sample outputs `phase5_outputs/` – 4 artefacts

Boldr's tickets vs. the wider market

SOURCES

r/Watches • WatchUSeek (microbrand forum) • Reddit r/AskWomenOver30 (gifter signal) • competitor review pages

BPA-free / health safety

INTERNAL

11/70 tickets • health_conscious is the #2 persona.

EXTERNAL

Watch forums treat BPA as a baseline expectation; rarely surfaces in reviews.

INSIGHT

Boldr-specific gap: customers ask because product pages don't surface it. Add a 'BPA-free' badge to listings — table stakes, not differentiator.

Sustainability / vegan straps

INTERNAL

Rising cluster; carbon-neutral shipping + vegan leather repeatedly flagged.

EXTERNAL

Microbrand forums show growing demand; Nordgreen and Solios lead on messaging.

INSIGHT

Market-wide concern, not Boldr-specific. Boldr risks falling behind — develop a vegan strap angle and a packaging-recyclable statement this quarter.

Nickel allergy / titanium safety

INTERNAL

Recurring across health_conscious tickets; mesh-bracelet trace nickel flagged.

EXTERNAL

Reddit r/Watches threads on titanium-only watches see steady traction.

INSIGHT

Boldr-specific opportunity: Grade 5 Ti is a true selling point. Lead with it on health-conscious landing pages — convert a passive feature into an acquisition message.

How this engine maps to the five criteria

25%

TECHNICAL EXECUTION

Working n8n flow, validated on a real ticket. 7-step intelligence loop, low-code, deployable today.

25%

SME IMPACT & BUSINESS VALUE

~30 CS hours recovered / month. 5 product-page gaps surfaced in one month. Loop compounds: gap rate decreases as KB grows.

20%

COST EFFICIENCY

Under SGD 50 / month all-in. Scales sub-linearly on free tiers. Commercially viable on Qwen pricing even without sponsor credits.

10%

RESPONSIBLE AI

Six explicit safeguards: human gates everywhere, no hallucination, audit trail, PII boundary, pricing safety, no auto-publish to KB.

20%

PRESENTATION QUALITY

Built by a 3-person CS team without engineers. Visual canvas any team lead can follow. Architecture documented in repo.

Thank you. Ready for questions.

Team AlphaBeta

Tejas Chavan · Ayush K Pachariwala

Repo: `github.com/tejasdotchavan/e27-boldr-challenge-AlphaBeta`

Live workflow: `tejasdotchavan.app.n8n.cloud`